

## Musterlösung Serie 10

### LINEAR MAPS

**1. Choice of basis.** Show that if  $V$  is a one-dimensional vector space over a field  $K$  and  $T \in \text{Hom}_K(V, V)$ , then there exists  $\lambda \in K$  such that for all  $v \in V : Tv = \lambda v$ . Explain then why an isomorphism  $V \rightarrow K$  depends on a choice of basis, while one from  $\text{Hom}_K(V, V)$  to  $K$  doesn't.

*Solution:* Let  $v_0 \in V \setminus \{0\}$ . Since  $V$  is one-dimensional,  $\{v_0\}$  is a basis of  $V$ . We have  $Tv_0 \in V$ , so there exists  $\lambda \in K$  such that  $Tv_0 = \lambda v_0$ . Now let  $v_1 \in V$ . For the same reason, there exists  $\mu \in K$  such that  $v_1 = \mu v_0$ . By linearity of  $T$ , we then have

$$Tv_1 = T(\mu v_0) = \mu T(v_0) = \mu \lambda v_0 = \lambda(\mu v_0) = \lambda v_1.$$

Since  $v_1$  was arbitrary, this proves the statement.

*Aliter:* You have seen in the lectures that such a linear map  $T$  can be represented by a matrix. Since a  $1 \times 1$  matrix is a scalar, this proves the statement.

Regarding the second part of the exercise, note that if we want to define a linear isomorphism from  $V$  to  $K$ , we need to first specify the image of a certain basis. Such a homomorphism is called “non-canonical”. On the other hand, we can define a linear isomorphism

$$\begin{array}{ccc} \text{Hom}(V, V) & \rightarrow & K \\ T & \mapsto & \lambda, \text{ where } \lambda \text{ is such that } \forall v \in V : Tv = \lambda v. \end{array}$$

In this case, we did not need to pick a basis in order to define the isomorphism. Such a homomorphism is called “canonical”.

**2. Linear map on a subspace.** Let  $V, W$  be vector spaces over a field  $K$ . Suppose that  $U \subsetneq V$  is a linear subspace and let  $S$  be a non-trivial element of  $\text{Hom}_K(U, W)$  (i.e. we assume that  $S$  does not map everything to 0). Define  $T : V \rightarrow W$  by

$$Tv = \begin{cases} Sv, & \text{if } v \in U \\ 0, & \text{if } v \in V \setminus U \end{cases}$$

Is  $T$  a linear map?

*Solution:* Let  $u \in U$  such that  $Su \neq 0$ . Such a  $u$  exists by our assumption on  $S$ . Let also  $v \in V \setminus U$  (since  $U \subsetneq V$ , this set is non-empty). We then observe that  $u+v \notin U$ . Indeed, if it were the case, then there would exist some  $u' \in U$  such that  $u+v = u' \Leftrightarrow v = u' - u \in U$ , which is a contradiction to our choice of  $v$ . Hence,

$$T(u+v) = 0 \neq Su + 0 = Tu + Tv,$$

which shows that  $T$  isn't linear.

**3. Linear map over a subspace.** Let  $U, V, W$  be vector spaces over a field  $K$  and let  $T : V \rightarrow W$  and  $S : W \rightarrow U$  be linear maps.

(a) Prove that

$$\text{rank}(S \circ T) \leq \min(\text{rank}(S), \text{rank}(T)).$$

(b) Show that  $\text{rank}(S \circ T) = \text{rank}(S)$  whenever  $T$  is surjective.

(c) Show that  $\text{rank}(S \circ T) = \text{rank}(T)$  whenever  $S$  is injective.

*Solution:*

(a) Note that since  $T$  is linear,  $\text{Im}(T)$  is a subspace of  $W$ . We define the restriction of  $S$  to  $\text{Im}(T)$  to be the linear map

$$\begin{aligned} S|_{\text{Im}(T)} : \text{Im}(T) &\rightarrow U \\ v &\mapsto S(v) \end{aligned}$$

Now note that  $\text{Im}(S \circ T) = \text{Im}(S|_{\text{Im}(T)}) \subset \text{Im}(S)$ . Hence

$$\text{rank}(S \circ T) = \text{rank}(S|_{\text{Im}(T)}) \leq \text{rank}(S). \quad (1)$$

On the other hand,

$$\text{rank}(T) = \dim(\text{Im}(T)) = \dim(\text{Ker}(S|_{\text{Im}(T)})) + \text{rank}(S|_{\text{Im}(T)}). \quad (2)$$

Therefore,

$$\text{rank}(S \circ T) = \text{rank}(S|_{\text{Im}(T)}) \leq \text{rank}(T).$$

This shows

$$\text{rank}(S \circ T) \leq \min\{\text{rank}(S), \text{rank}(T)\}.$$

(b) If  $T$  is surjective, the subspace  $\text{Im}(T)$  is now the whole of  $W$ . Then

$$S|_{\text{Im}(T)} = S|_W = S$$

and by (1), we have

$$\text{rank}(S \circ T) = \text{rank}(S|_{\text{Im}(T)}) = \text{rank}(S).$$

(c) If  $S$  is injective, we have  $\dim(\text{Ker}(S)) = 0$ , so  $\dim(\text{Ker}(S|_{\text{Im}(T)})) = 0$ . Hence, by (2), we obtain

$$\text{rank}(T) = \text{rank}(S|_{\text{Im}(T)}) = \text{rank}(S \circ T).$$

**4. Projection.** Let  $V$  be a vector space. An endomorphism  $P : V \rightarrow V$  satisfying  $P^2 := P \circ P = P$  is called idempotent or a projection. Show:

- (a) For every projection  $P$ , its image  $\text{Im}(P)$  is a linear complement of  $\text{Ker}(P)$  in  $V$ .
- (b) For any subvectorspaces  $W_1, W_2 \subset V$ , such that  $W_1$  is a complement of  $W_2$  in  $V$ , there exists a unique projection  $P : V \rightarrow V$  with

$$\text{Ker}(P) = W_1 \quad \text{und} \quad \text{Im}(P) = W_2.$$

*Solution:*

- (a) Let  $P : V \rightarrow V$  be a projection and define

$$W_1 := \text{Ker}(P) \quad \text{and} \quad W_2 := \text{Im}(P).$$

We need to show that  $W_1 + W_2 = V$  and  $W_1 \cap W_2 = \{0\}$ .

For any  $v \in V$  we have

$$P(v - P(v)) = P(v) - P^2(v) = P(v) - P(v) = 0,$$

hence  $v - P(v) \in \text{Ker}(P) = W_1$ . Moreover, we have  $P(v) \in \text{Im}(P) = W_2$ . Thus, we get

$$v = (v - P(v)) + P(v) \in W_1 + W_2,$$

and  $W_1 + W_2 = V$ .

Now let  $v \in W_1 \cap W_2$ . Since  $v$  lies in the image of  $P$ , there exists a  $w \in V$  with  $P(w) = v$ . Application of  $P$  on both sides of the equation yields  $v = P(w) = P^2(w) = P(v)$ . As  $v$  is also contained in the kernel of  $P$ , we have  $v = P(v) = 0$ . Thus, we get  $W_1 \cap W_2 = \{0\}$ .

- (b) Let  $W_1, W_2 \subseteq V$  be any subvectorspaces of  $V$  satisfying  $V = W_1 + W_2$  and  $W_1 \cap W_2 = \{0\}$ . For any  $v \in V$  there exist unique  $w_1 \in W_1$  and  $w_2 \in W_2$  such that we have  $v = w_1 + w_2$ . Consider the map  $P: V \rightarrow V$ , which maps  $v \in V$  to this  $w_2 \in W_2$ . One can show by explicit computations that  $P$  is linear and a projection with  $W_1 = \text{Ker}(P)$  and  $W_2 = \text{Im}(P)$ . We leave this to the reader.

It remains to show that  $P$  is unique. Let  $P'$  be another projection with  $W_1 = \text{Ker}(P')$  and  $W_2 = \text{Im}(P')$ . Then we have

$$P|_{W_1} = 0 = P'|_{W_1}.$$

For an arbitrary element  $v \in W_2$  there exist  $w, w' \in V$  such that  $P(w) = v$  and  $P'(w') = v$ . This yields

$$P(v) - P'(v) = P(P(w)) - P'(P'(w')) = P(w) - P'(w') = v - v = 0$$

and hence we get  $P|_{W_2} = P'|_{W_2}$ . As we have  $V = W_1 + W_2$  it follows that  $P = P'$ .

**5. Preimage of a linear map.** Let  $f: V \rightarrow W$  be a linear map of  $K$ -vector spaces. Show:

- (a) For every subvectorspace  $W' \subset W$  the preimage

$$f^{-1}(W') := \{v \in V \mid f(v) \in W'\}$$

is a subvectorspace of  $V$ .

- (b) We have

$$\dim f^{-1}(W') = \dim \text{Ker}(f) + \dim (\text{Im}(f) \cap W').$$

*Solution:* Let  $V' := f^{-1}(W')$ .

- (a) We need to show that  $V'$  is non empty and that for arbitrary  $x, y \in V'$  and  $\alpha \in K$  the sum  $x + y$  and the product  $\alpha x$  is again contained in  $V'$ .

As  $f(0) = 0 \in W'$ , the vector  $0$  is contained in  $V'$  and thus  $V'$  is non empty. Linearity of  $f$  yields

$$f(x + y) = f(x) + f(y) \quad \text{and} \quad f(\alpha x) = \alpha f(x).$$

As  $f(x), f(y) \in W'$ , we get from the axioms of subvectorspaces that  $f(x) + f(y)$  and  $\alpha f(x)$  are again contained in  $W'$ . Therefore, we have that  $x + y$  and  $\alpha x$  are contained in  $V'$ .

*Note.* Is it also the case for the image of a linear subspace via a linear map?

(b) The definition of  $V'$  yields that there exists a map

$$f': V' \rightarrow W', \quad v' \mapsto f'(v') := f(v).$$

As  $f$  is linear, the newly defined  $f'$  is linear as well. We get

$$\dim(V') = \dim(\text{Ker}(f')) + \dim(\text{Im}(f')). \quad (3)$$

Since  $0 \in W'$ , it follows that  $\text{Ker}(f) \subset V'$ . Plugging in the definition, we get  $\text{Ker}(f') = \text{Ker}(f)$ . Moreover, we have

$$\begin{aligned} \text{Im}(f') &= \{w \in W' \mid \exists v \in V' : f'(v) = w\} \\ &= \{w \in W' \mid \exists v \in V : f(v) = w\} \\ &= \{w \in W \mid \exists v \in V : f(v) = w \text{ und } w \in W'\} \\ &= \text{Im}(f) \cap W'. \end{aligned}$$

Plugging this into (3), we get

$$\dim(V') = \dim(\text{Ker}(f)) + \dim(\text{Im}(f) \cap W').$$

**6. Isomorphisms of vector spaces.** Let  $V, W$  be vector spaces over a field  $K$  and let  $T : V \rightarrow W$  be an isomorphism of vector spaces. Show that:

- (a)  $T$  maps linearly independent sets to linearly independent sets;
- (b)  $T$  maps spanning sets of  $V$  to spanning sets of  $W$ ;
- (c)  $T$  maps bases to bases.

*Lösung:*

- (a) Assume that  $\{v_i \mid 1 \leq i \leq n\}$  is linearly independent. We prove that  $\{T(v_i)\}$  is linearly independent. Assume that there exists  $\{a_i \mid 1 \leq i \leq n\} \subset K$  such that

$$\sum_{1 \leq i \leq n} a_i T(v_i) = 0 \Leftrightarrow T \left( \sum_{1 \leq i \leq n} a_i v_i \right) = 0.$$

This is equivalent to  $\sum_{1 \leq i \leq n} a_i v_i = 0$  since  $T$  is an isomorphism. Since we assumed  $\{v_i \mid 1 \leq i \leq n\}$  to be linearly independent, this is equivalent to  $\forall i : a_i = 0$ . Summing up, we have

$$\sum_{1 \leq i \leq n} a_i T(v_i) = 0 \Leftrightarrow \forall 1 \leq i \leq n : a_i = 0.$$

Hence,  $\{T(v_i) \mid 1 \leq i \leq n\}$  is linearly independent.

- (b) Assume that  $S = \{v_1, v_2, \dots, v_r\}$  is a spanning set of  $V$ . Let  $w$  be an arbitrary element of  $W$ . Since  $T$  is an isomorphism,  $w$  admits a unique preimage  $v$  in  $V$ . Since  $V = \text{Sp}(v_1, \dots, v_r)$ , we can write  $v$  uniquely as  $v = \sum_{1 \leq i \leq r} a_i v_i$  for  $\{a_i \mid 1 \leq i \leq r\} \subset K$ . Therefore,

$$w = T(v) = T\left(\sum_{1 \leq i \leq r} a_i v_i\right) = \sum_{1 \leq i \leq r} a_i T(v_i)$$

Since  $w$  was arbitrary, this shows  $W = \text{Sp}(\{T(v_i) \mid 1 \leq i \leq r\})$ .

- (c) Since a basis is a linearly independent spanning subset. The combination of (a) and (b) shows (c).

**7. Dimension and basis.** Let  $V, W$  be vector spaces over  $\mathbb{Q}$ . We say that a map  $f : V \rightarrow W$  is additive, if

$$\forall x \in V \forall y \in V : f(x + y) = f(x) + f(y).$$

Show that

$$\text{Hom}_{\mathbb{Q}}(V, W) = \{f : V \rightarrow W \mid f \text{ is additive}\}.$$

*Solutions:* We want to show that any additive map  $f : V \rightarrow W$  also satisfies  $\forall q \in \mathbb{Q}, \forall v \in V : f(qv) = qf(v)$ . Let  $v \in V$ . We first note that

$$f(0) = f(0 + 0) = f(0) + f(0) \implies f(0) = 0.$$

It directly follows that  $0 = f(v + (-v)) = f(v) + f(-v)$ . Hence,  $f(-v)$  is the additive inverse of  $f(v)$  in  $W$ .

We continue by showing by induction that for any  $m \in \mathbb{N}$ ,  $f(mv) = mf(v)$ . The above shows the base step  $m = 0$ . We now assume that it is proved for all  $m$  with  $0 \leq m \leq k$  and we prove it for  $k + 1$ . We have

$$f((k + 1)v) = f(kv + v) = f(kv) + f(v) = kf(v) + f(v) = (k + 1)f(v),$$

where we used the induction hypothesis to obtain the second-to-last equality. Now, for any negative integer  $n \in \mathbb{Z}$ , we do have  $-n \in \mathbb{N}$ . So,

$$f(nv) = f((-n)(-v)) = (-n)f(-v) = (-n)(-f(v)) = nf(v).$$

We have shown that  $f(nv) = nf(v)$  for all  $n \in \mathbb{Z}$ .

Finally, notice that for any  $n \in \mathbb{Z} \setminus \{0\}$ ,

$$\frac{n}{n}f(v) = f(v) = f\left(\frac{n}{n}v\right) = nf\left(\frac{1}{n}v\right) \Leftrightarrow n\left(\frac{1}{n}f(v) - f\left(\frac{1}{n}v\right)\right) = 0.$$

This is equivalent to  $\frac{1}{n}f(v) = f\left(\frac{1}{n}v\right)$ .

We conclude that for all  $m \in \mathbb{Z}$ , for all  $n \in \mathbb{Z} \setminus \{0\}$ , for all  $v \in V$ :

$$f\left(\frac{m}{n}v\right) = mf\left(\frac{1}{n}v\right) = \frac{m}{n}f(v).$$

Since any  $q \in \mathbb{Q}$  can be written as  $q = \frac{m}{n}$  for some  $m \in \mathbb{Z}$  and some  $n \in \mathbb{Z} \setminus \{0\}$ , we have proved that  $f$  is  $\mathbb{Q}$ -linear.